

A Convex-Operator Envelope Theorem for Dedekind Complete Riesz-Valued Maps

Solution packet for arXiv:2404.10900

July 3, 2026

Status

Result type: candidate full solution, pending human review.

Source paper. Mario Ghossoub, Giulio Principi, and Ruodu Wang, *Allocation Mechanisms in Decentralized Exchange Markets with Frictions*, arXiv:2404.10900.

Target. Section 6 of the source paper states that the authors prove order-theoretic envelope representations for sublinear operators with values in Dedekind complete Riesz spaces, and then says that “the generalization of our results toward convex operators remains an open question.” The statement below answers the direct finite-valued algebraic version of that question.

6. A DISCUSSION OF THE MATHEMATICAL CONTRIBUTION

The results presented here on the axiomatization of allocation mechanisms are based on applications of Hahn-Banach type-extension results (Theorems [A.1](#) and [A.2](#)). These results provide conditions for extending linear operators on ordered vector spaces. We study some implications of these extension results for envelope representations of sublinear operators. As in the standard real-valued setting, we show that sublinear operators can be written as the upper envelopes of their dominated linear operators. Our results abstract from any topological assumptions, and they are purely order-theoretic and algebraic in nature. The level of generality adopted makes our results applicable to all order-theoretic contexts where sublinear operators play a role. In particular, we retrieve such envelope representation results within Dedekind complete Riesz spaces, that include a large amount of widely used ordered vector spaces. The provided representation results are then refined, adding further restrictions on the behavior of the sublinear operators, and they highlight a classic feature: the dominated linear operators often inherit the same properties of the dominating sublinear operator. Among such restrictions, the main ones are monotonicity and normalization, that are both meaningful in economics and finance. Monotonicity is a non-satiation assumption of the type “*more is better*”, while normalization says that the value of certain outcomes should remain unchanged after any evaluation. This motivates the mathematical analysis of such properties, that is provided in the [Appendices](#). To the best of our knowledge, the generalization of our results toward convex operators remains an open question.

From a mathematical perspective, our representations results take place in the literature on

Open Problem Transcription

The source paper writes:

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Definitions

Let V be a real vector space and let $(W, \succeq)$ be a partially ordered real vector space. A map $f : V \rightarrow W$ is $\succeq$ -convex if

$$f(\lambda x + (1 - \lambda)y) \preceq \lambda f(x) + (1 - \lambda)f(y)$$

for all $x, y \in V$ and all $\lambda \in [0, 1]$.

An affine map $a : V \rightarrow W$ is a map of the form

$$a(x) = Lx + c, \quad L \in L(V, W), \quad c \in W.$$

For two maps $a, f : V \rightarrow W$, write $a \preceq f$ when $a(x) \preceq f(x)$ for every $x \in V$. Define

$$\text{Aff}(f) = \{a : V \rightarrow W : a \text{ is affine and } a \preceq f\}.$$

If V is also partially ordered by $\succeq_V$, let $\text{Aff}_+(f)$ denote the members of $\text{Aff}(f)$ whose linear part is $(\succeq_V, \succeq)$ -positive.

Proof Intuition

For a convex real-valued function, a supporting affine minorant at x can be obtained from a subgradient at x . The same idea works here, but the subgradient is produced algebraically rather than topologically. At a base point x , take the order-valued directional derivative

$$p_x(y) = \inf_{t>0} \frac{f(x + ty) - f(x)}{t}.$$

Convexity makes these quotients decreasing as $t \downarrow 0$ and also gives a lower bound, so the infimum exists in the Dedekind complete Riesz space W . The function p_x is sublinear. The Hahn-Banach extension theorem quoted in the source paper therefore gives a linear $L_x \preceq p_x$. Since $p_x(y) \preceq f(x + y) - f(x)$, the map

$$a_x(z) = f(x) + L_x(z - x)$$

is an affine minorant of f touching f at x . Taking one such minorant for each base point gives the desired envelope representation. If f is monotone, then p_x is monotone, and the Hahn-Banach-Kantorovich variant gives L_x positive.

Main Result

Theorem 1 (Affine envelope for order-convex maps). *Let V be a real vector space and let $(W, \succeq)$ be a Dedekind complete Riesz space. If $f : V \rightarrow W$ is $\succeq$ -convex, then*

$$f(x) = \sup_{a \in \text{Aff}(f)} a(x), \quad x \in V.$$

The supremum is the lattice supremum in W .

If, in addition, $(V, \succeq_V)$ is a partially ordered vector space and f is $(\succeq_V, \succeq)$ -monotone, then

$$f(x) = \sup_{a \in \text{Aff}_+(f)} a(x), \quad x \in V.$$

Lemma 1 (Directional quotient). *Fix $x \in V$ and define*

$$q_t(y) = \frac{f(x + ty) - f(x)}{t}, \quad t > 0, y \in V.$$

Then, for each $y \in V$, the family $\{q_t(y) : t > 0\}$ is bounded below and has an infimum

$$p_x(y) := \inf_{t > 0} q_t(y)$$

in W . The map $p_x : V \rightarrow W$ is $\succeq$ -sublinear, i.e.

$$p_x(y) + p_x(z) \succeq p_x(y + z), \quad p_x(\alpha y) = \alpha p_x(y) \quad (\alpha \geq 0).$$

If V is ordered and f is monotone, then p_x is monotone.

Proof. If $0 < s < t$, then

$$x + sy = \left(1 - \frac{s}{t}\right)x + \frac{s}{t}(x + ty).$$

Convexity gives

$$f(x + sy) \preceq \left(1 - \frac{s}{t}\right)f(x) + \frac{s}{t}f(x + ty),$$

and hence $q_s(y) \preceq q_t(y)$. Thus the quotients are increasing in the ordinary parameter t , or decreasing as $t \downarrow 0$.

They are bounded below. Indeed, for $t > 0$,

$$x = \frac{1}{1+t}(x + ty) + \frac{t}{1+t}(x - y).$$

Convexity yields

$$f(x) \preceq \frac{1}{1+t}f(x + ty) + \frac{t}{1+t}f(x - y),$$

which rearranges to

$$f(x) - f(x - y) \preceq q_t(y).$$

Since W is a Dedekind complete Riesz space, lower bounded nonempty subsets have infima. Hence $p_x(y)$ is well-defined.

Positive homogeneity is immediate for $\alpha > 0$ from

$$q_t(\alpha y) = \alpha q_{\alpha t}(y),$$

and the case $\alpha = 0$ follows from $q_t(0) = 0$.

For subadditivity, convexity gives

$$x + t(y + z) = \frac{1}{2}(x + 2ty) + \frac{1}{2}(x + 2tz),$$

so

$$q_t(y + z) \preceq q_{2t}(y) + q_{2t}(z), \quad t > 0.$$

It remains only to pass to infima. We use the elementary order fact that if $a_t \downarrow a$ and $b_t \downarrow b$ over the same directed parameter set, then $a_t + b_t \downarrow a + b$. To verify it, let c be a lower bound of $a_t + b_t$. For any two parameters s, u , choose t below both. Then $c \preceq a_s + b_u$. Taking the infimum first over s and then over u gives $c \preceq a + b$. Applying this to $a_t = q_t(y)$ and $b_t = q_t(z)$ gives

$$p_x(y + z) \preceq \inf_{t > 0} (q_{2t}(y) + q_{2t}(z)) = p_x(y) + p_x(z).$$

Finally, if f is monotone and $y \succeq_V z$, then $x + ty \succeq_V x + tz$ for all $t > 0$, hence $q_t(y) \succeq q_t(z)$ for all $t > 0$. Taking infima gives $p_x(y) \succeq p_x(z)$. $\square$

Proof of Theorem 1. Fix $x \in V$ and form p_x as in Lemma 1. By Theorem A.1 of the source paper, applied to the zero subspace and the zero linear map, there is a linear $L_x : V \rightarrow W$ such that

$$L_x(y) \preceq p_x(y), \quad y \in V.$$

For any $z \in V$, putting $y = z - x$ and using the term $t = 1$ in the infimum gives

$$L_x(z - x) \preceq p_x(z - x) \preceq f(z) - f(x).$$

Therefore

$$a_x(z) := f(x) + L_x(z - x)$$

is an affine map satisfying $a_x \preceq f$, and it touches f at x : $a_x(x) = f(x)$.

For each fixed z , the set $\{a(z) : a \in \text{Aff}(f)\}$ is nonempty and bounded above by $f(z)$, so its supremum exists in W . Since every $a \in \text{Aff}(f)$ is dominated by f , we have

$$\sup_{a \in \text{Aff}(f)} a(z) \preceq f(z).$$

Taking the touching minorant a_z constructed at z gives the reverse inequality, because $a_z(z) = f(z)$. Hence

$$f(z) = \sup_{a \in \text{Aff}(f)} a(z).$$

If V is ordered and f is monotone, Lemma 1 shows that p_x is monotone. Applying Theorem A.2 of the source paper instead of Theorem A.1 gives L_x positive. The same touching-minorant argument then gives the representation over $\text{Aff}_+(f)$. $\square$

Corollary 1 (Largest representing family). *The family $\text{Aff}(f)$ is convex and is the largest family of affine minorants whose pointwise lattice supremum represents f . In the monotone case, $\text{Aff}_+(f)$ has the corresponding largest-family property among affine minorants with positive linear part.*

Proof. Convexity of $\text{Aff}(f)$ is immediate: if $a, b \preceq f$ and $\lambda \in [0, 1]$, then $\lambda a + (1 - \lambda)b \preceq f$. If a family $\mathcal{A}$ of affine maps represents f by a pointwise supremum, then each $a \in \mathcal{A}$ satisfies $a(x) \preceq f(x)$ for all x , hence $\mathcal{A} \subseteq \text{Aff}(f)$. The same proof applies to $\text{Aff}_+(f)$. $\square$

Verification Notes

The source paper's Appendix A proves the Hahn-Banach extension theorem (Theorem A.1) and the positive Hahn-Banach-Kantorovich variant (Theorem A.2) for maps into Dedekind complete Riesz spaces. Those are the only external mathematical tools used here.

The theorem is finite-valued, matching the source paper’s maps $h : V \rightarrow W$. It does not address proper extended-valued convex operators, topological lower-semicontinuity issues, or conditional-expectation penalty representations. It does, however, give the direct affine-minorant envelope generalization of the source paper’s sublinear envelope theorem and its monotone-positive refinement.

No computational verification is relevant. The packet was rendered from this LaTeX source, and the source paper PDF was rebuilt locally from the arXiv source bundle. The open-question crop is from page 22 of that rebuilt PDF.

Novelty and Search Bounds

Local run indexes were searched for arXiv id 2404.10900, the source title, `convex operators`, `order-convex`, `affine minorants`, `affine envelope`, `Hahn-Banach-Kantorovich`, and `Dedekind complete Riesz`. No prior run packet or attempt was found for this exact target before this packet was created.

Bounded web searches on July 3, 2026 for exact and close phrases around the source sentence and the theorem statement found no matching answer. Scalar supporting-affine-minorant theorems are classical; this packet claims the order-valued finite-valued theorem prompted by the source paper, not a complete survey of all vector-valued convex duality.

Human Review Recommendation

Check Lemma 1, especially the lower boundedness of the quotient family and the infimum-of-sums step in the Riesz order. If those steps are accepted, the rest is a direct application of the source paper’s Hahn-Banach-Kantorovich theorem and should be considered a full finite-valued answer to the source paper’s stated convex-operator generalization question.

References

- [1] M. Ghossoub, G. Principi, and R. Wang, *Allocation Mechanisms in Decentralized Exchange Markets with Frictions*, arXiv:2404.10900.
- [2] C. D. Aliprantis and O. Burkinshaw, *Positive Operators*, Springer, 2006.